

GENERATIVE AI VOCABULARY

1000+ Terms Every AI Learner Should Know

By: Nitish Raj

Gen AI

LLM Fundamentals

Transformer Architecture

- Self-Attention / Multi-Head Attention
- Positional Encoding
 - Sinusoidal (Original)
 - RoPE (Rotary Position Embedding)
 - ALiBi (Attention with Linear Biases)
- Feed-Forward Network (FFN)
- Layer Normalization
 - Pre-Norm
 - Post-Norm
- Residual Connections
- Architecture Variants
 - Encoder-Only (BERT, RoBERTa, DeBERTa)
 - Decoder-Only (GPT, LLaMA, Claude, Mistral)
 - Encoder-Decoder (T5, BART, Flan-T5)

Tokenization

- BPE (Byte-Pair Encoding)
- WordPiece
- SentencePiece
- Tiktoken
- Token
 - Input Tokens
 - Output Tokens
 - Special Tokens ([CLS], [SEP], <|endoftext|>)

Context Window

- Short Context (4K–8K)
- Long Context (32K–128K)
- Ultra-Long Context (200K+)
- Context Window Extension
 - Sliding Window Attention
 - Sparse Attention
 - Ring Attention

Decoding / Generation

- Autoregressive Decoding
- Temperature
- Top-k Sampling
- Top-p (Nucleus) Sampling

- Beam Search
- Greedy Decoding
- Repetition Penalty
- Stop Sequences
- Frequency Penalty
- Presence Penalty
- Speculative Decoding

— **Key Concepts**

- Pre-training
- Next Token Prediction
- Masked Language Modeling (MLM)
- Perplexity
- Cross-Entropy Loss
- Softmax
- Logits
- Log-Probabilities
- Vocabulary Size

— **Multimodal**

- Vision-Language Models (VLMs)
- Text-to-Image
- Text-to-Audio
- Text-to-Video
- Image Understanding
- Audio Understanding (Speech-to-Text)
- Omni Models (GPT-4o, Gemini)

— **NLP (Natural Language Processing)**

- Text Classification
- Named Entity Recognition (NER)
- Sentiment Analysis
- Machine Translation
- Summarization
 - Extractive
 - Abstractive
- Question Answering
- Text Generation
- Coreference Resolution
- Part-of-Speech Tagging
- Dependency Parsing
- Semantic Similarity

- Intent Detection
- Slot Filling
- Information Extraction

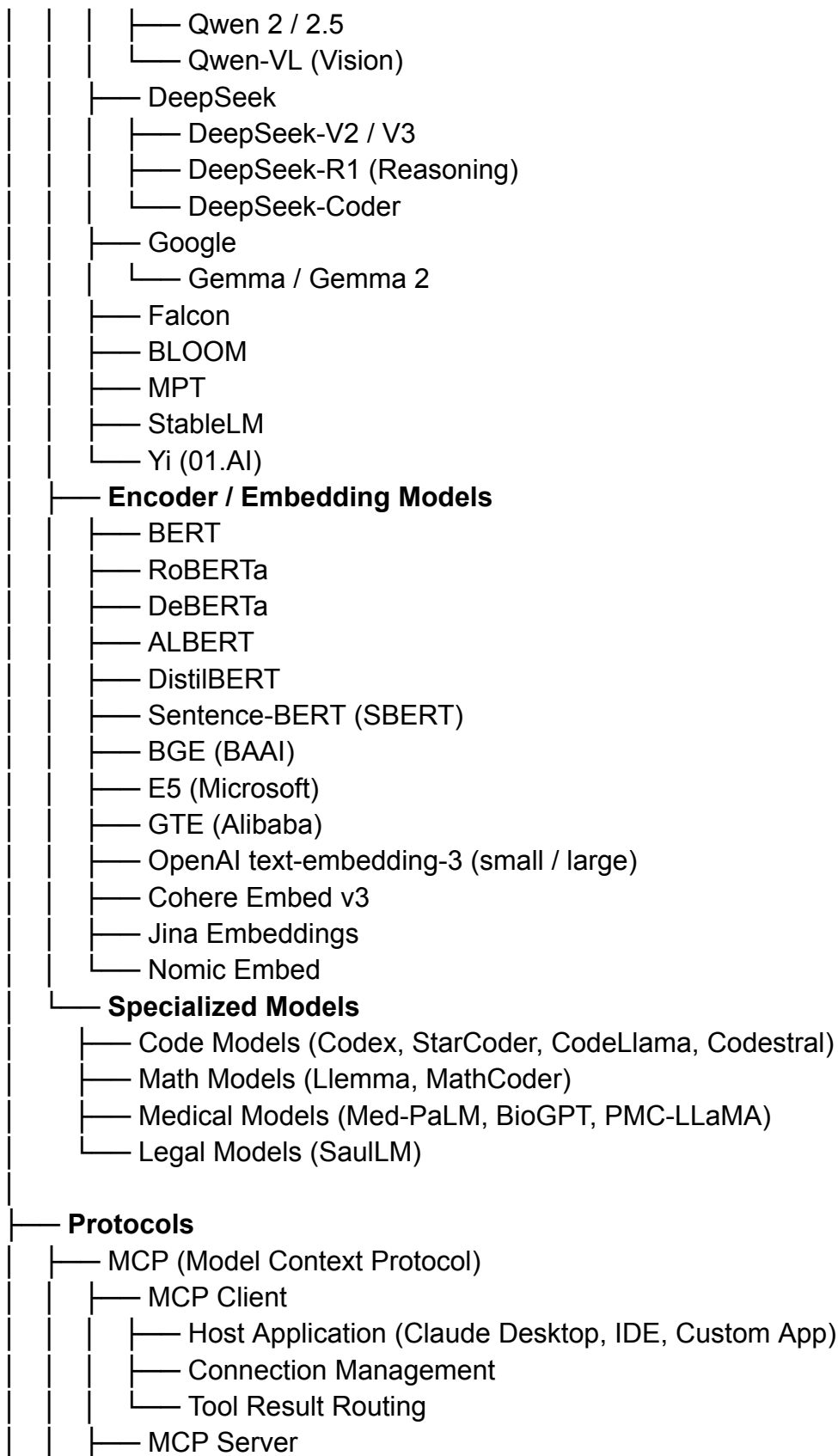
Models & Providers

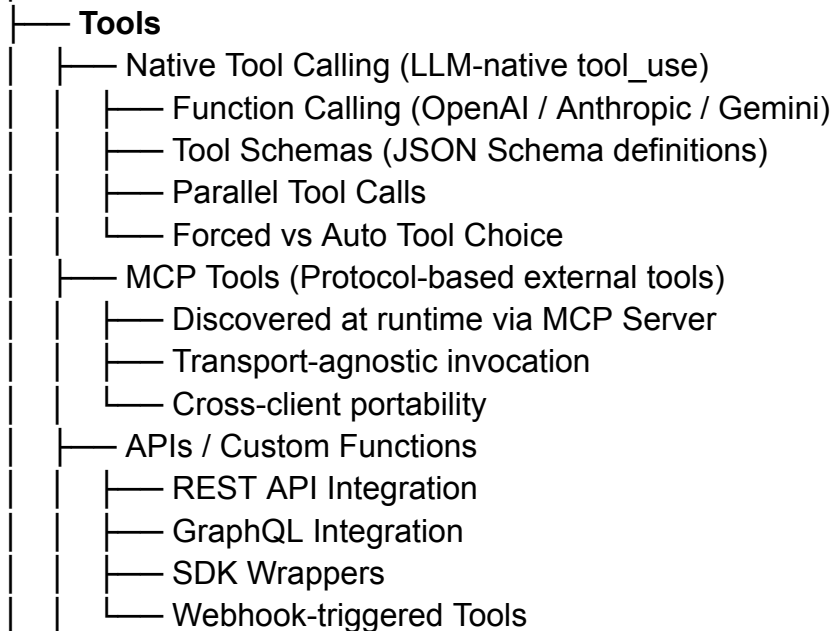
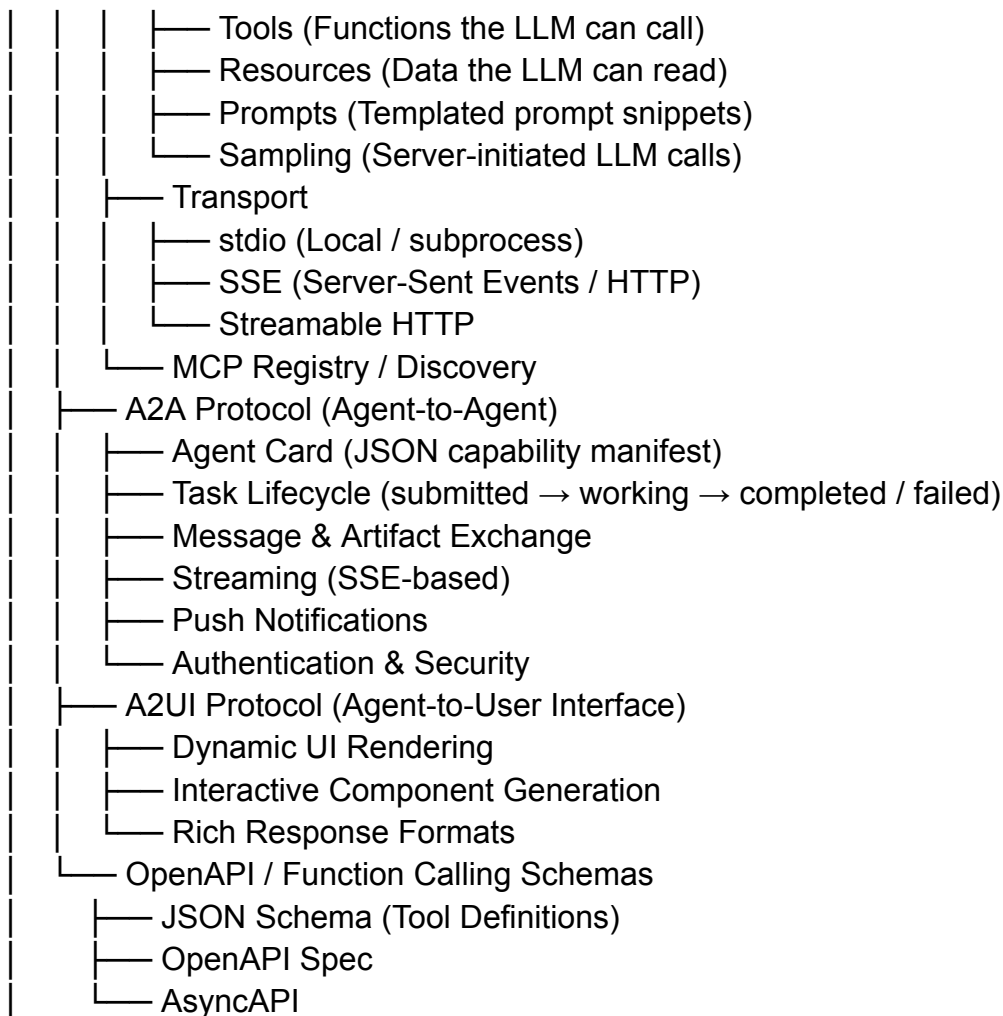
Closed-Source / API-Only

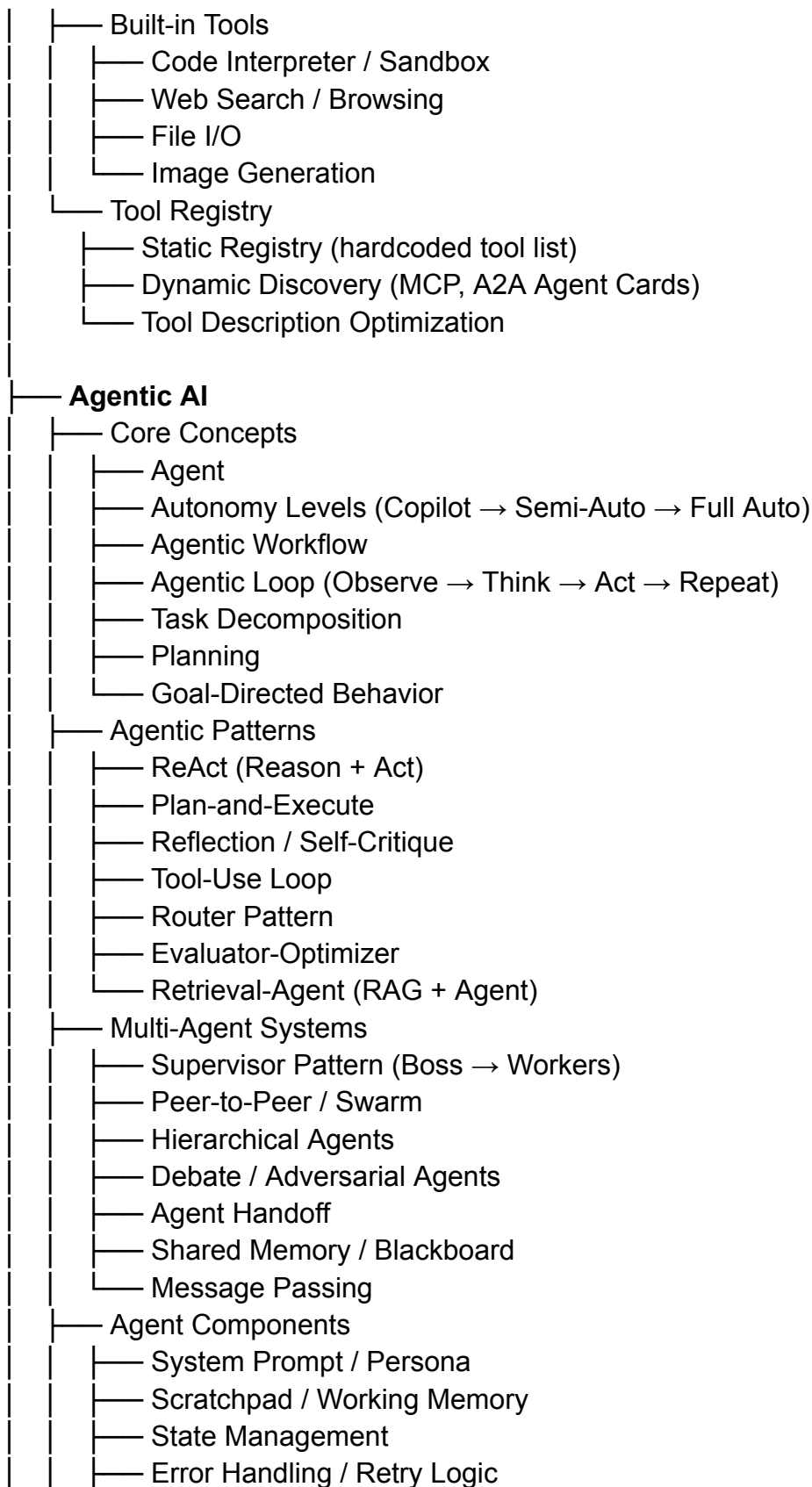
- OpenAI
 - GPT-4 / GPT-4o / GPT-4o-mini
 - o1 / o3 / o4-mini (Reasoning Models)
 - DALL·E (Image Generation)
 - Whisper (Speech-to-Text)
 - TTS (Text-to-Speech)
 - Codex (Legacy)
 - Assistants API
- Anthropic / Claude
 - Claude Opus
 - Claude Sonnet
 - Claude Haiku
 - Constitutional AI (Training Method)
- Google / Gemini
 - Gemini Ultra / Pro / Flash / Nano
 - PaLM (Legacy)
- Cohere
 - Command R / R+
 - Embed v3
 - Rerank

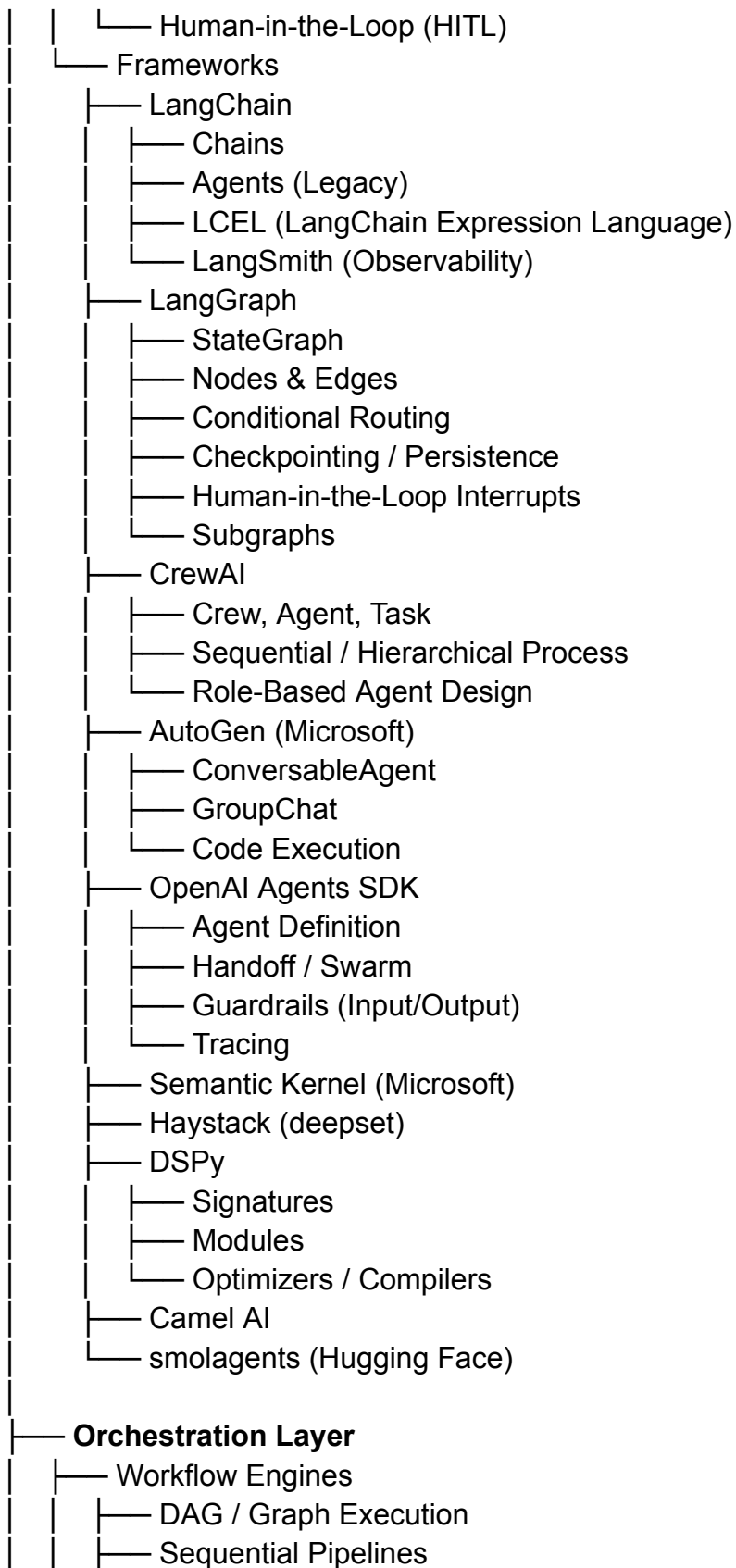
Open Weights

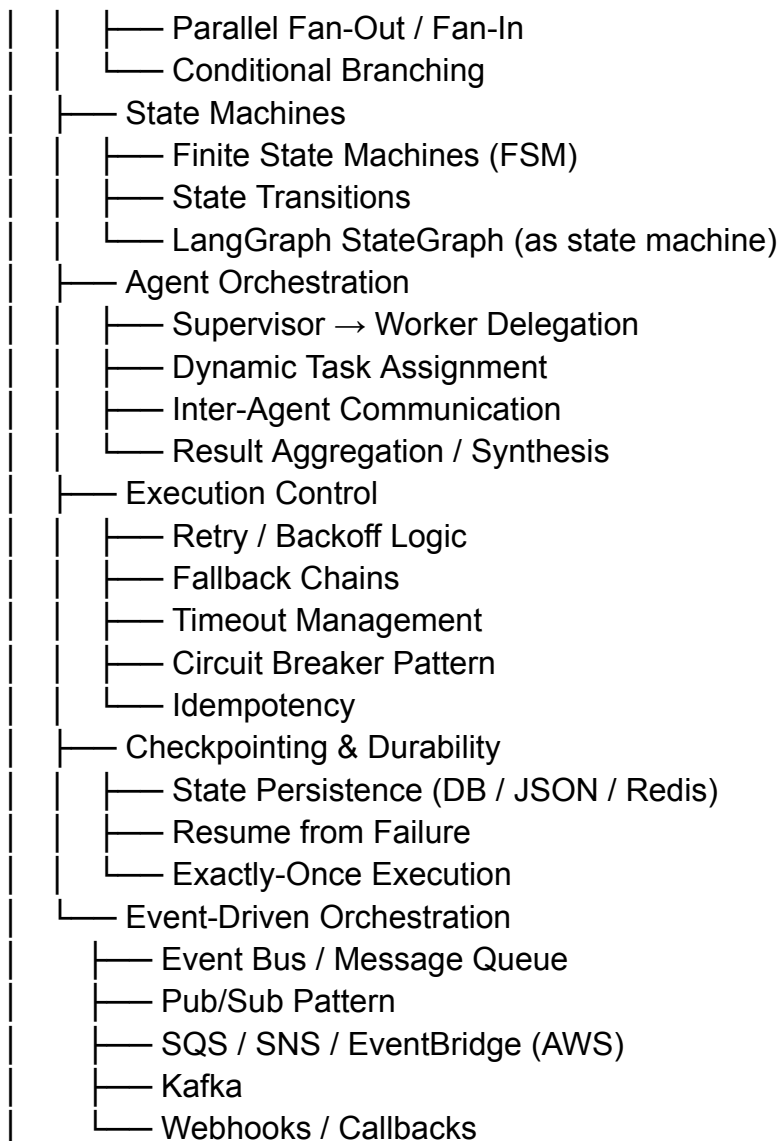
- Meta / LLaMA
 - LLaMA 2
 - LLaMA 3 / 3.1 / 3.2 / 4
 - Code LLaMA
- Mistral AI
 - Mistral 7B
 - Mixtral 8x7B / 8x22B (MoE)
 - Mistral Large / Medium / Small
 - Codestral
- Microsoft / Phi
 - Phi-2 / Phi-3 / Phi-4
 - Phi-Silica (On-Device)
- Alibaba / Qwen



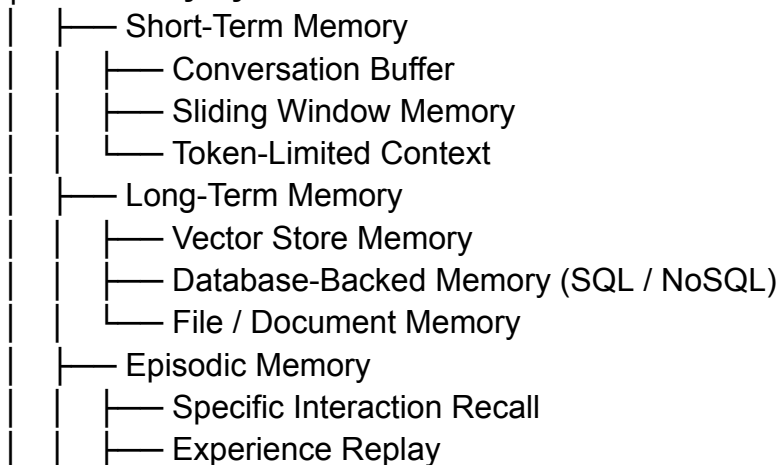


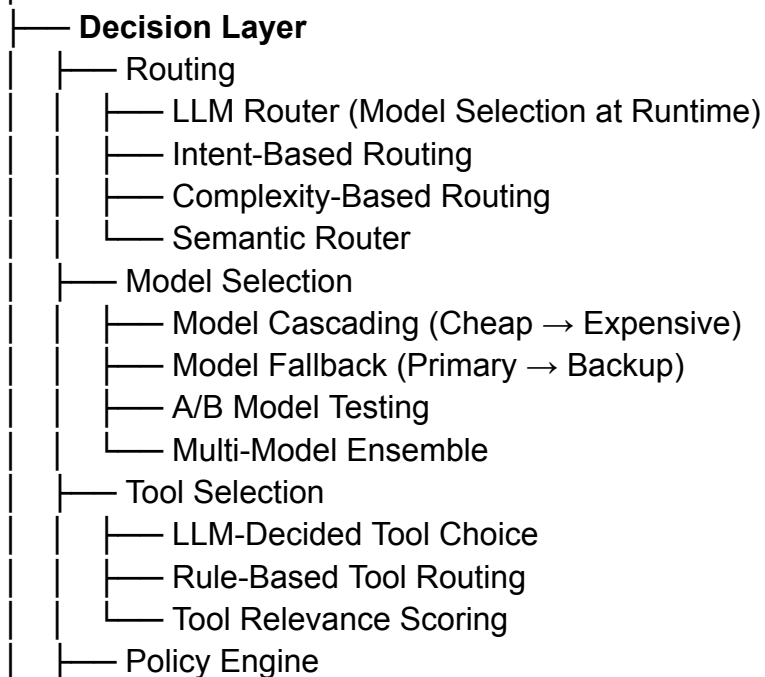
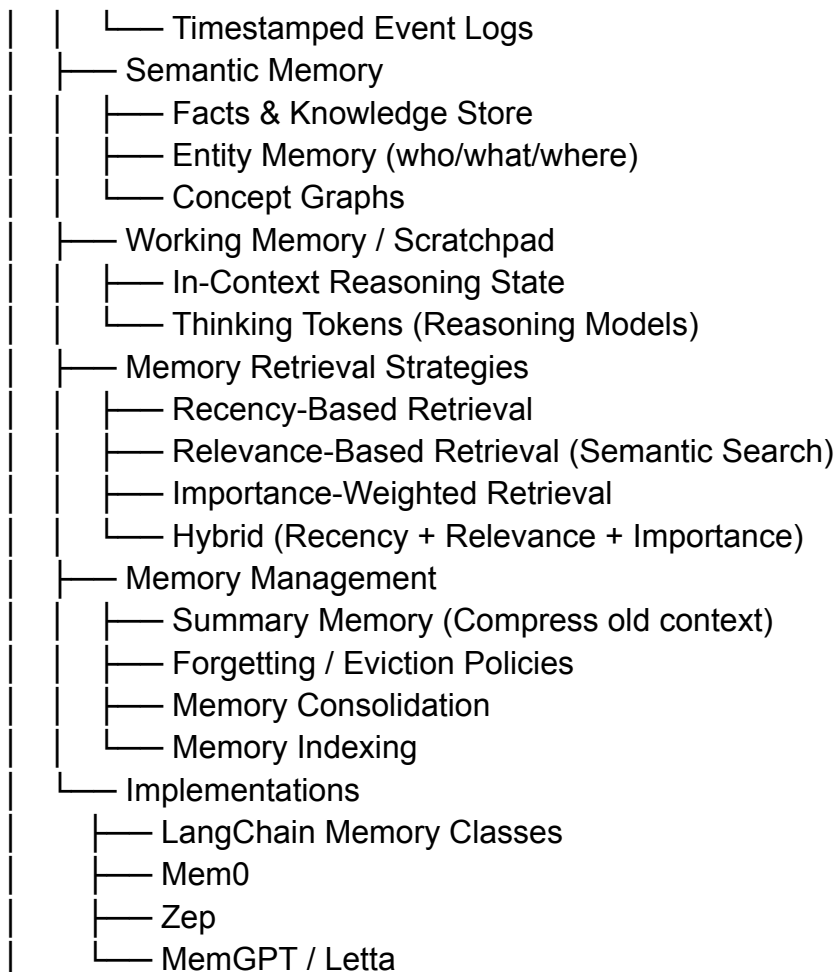


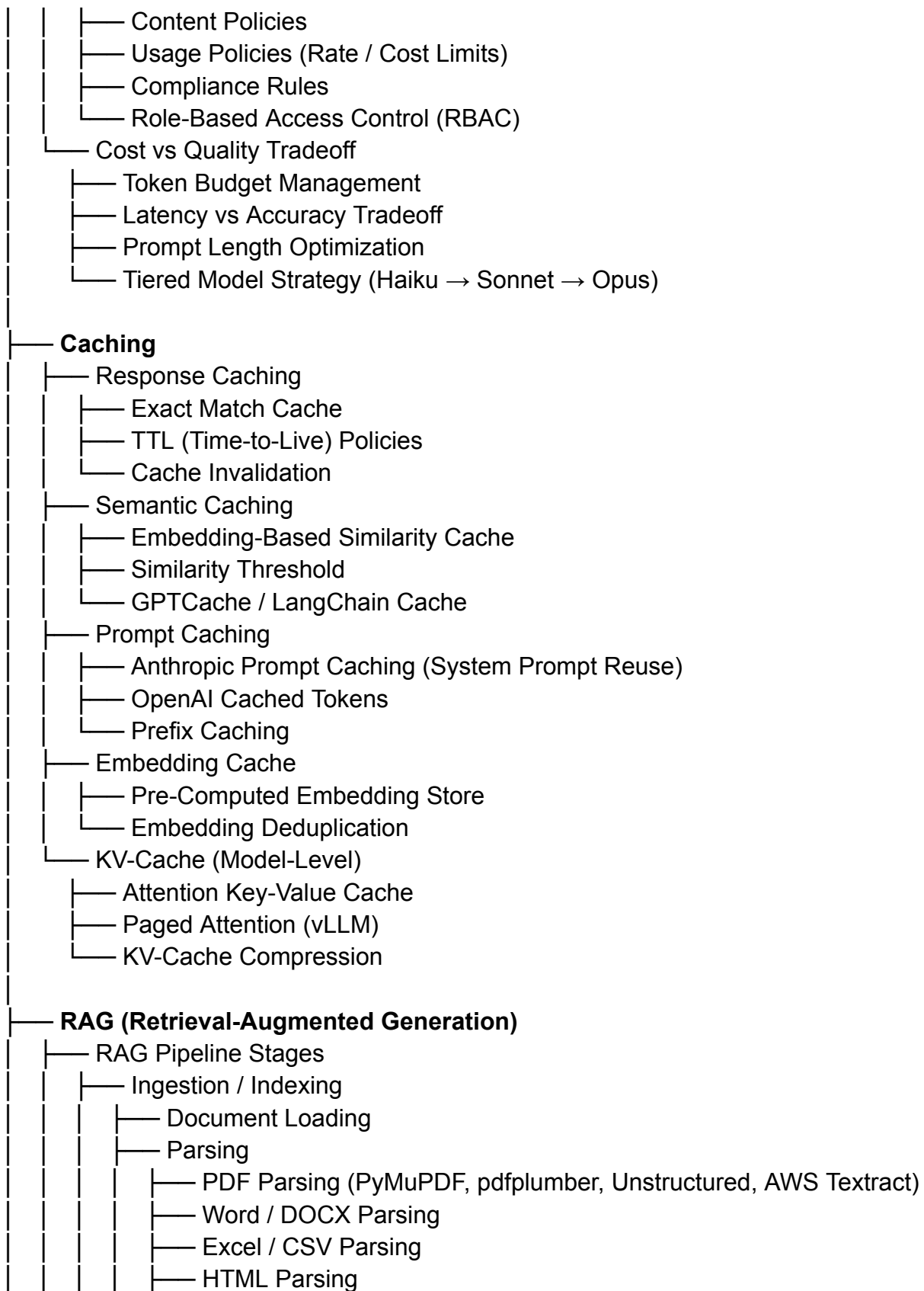


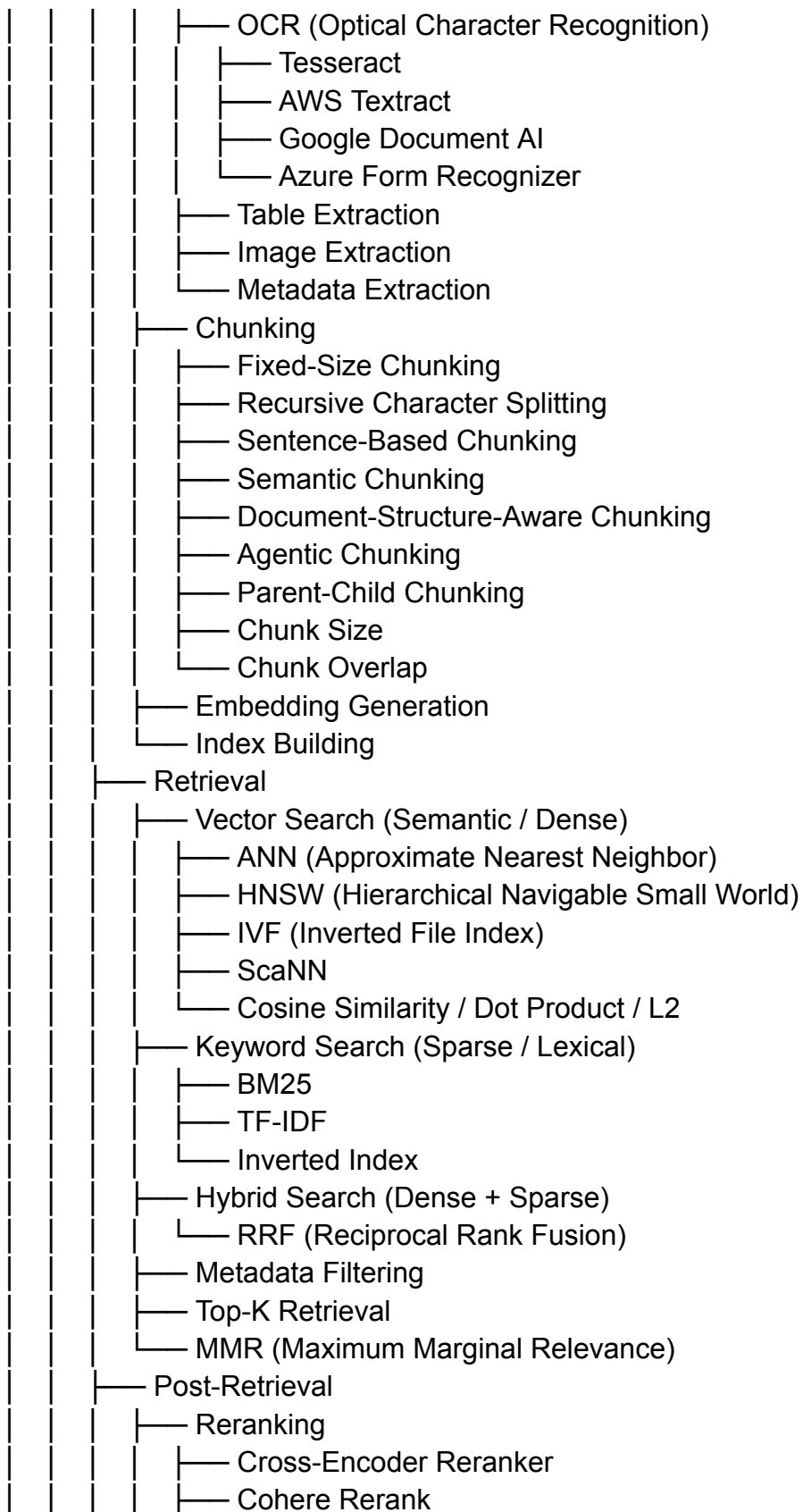


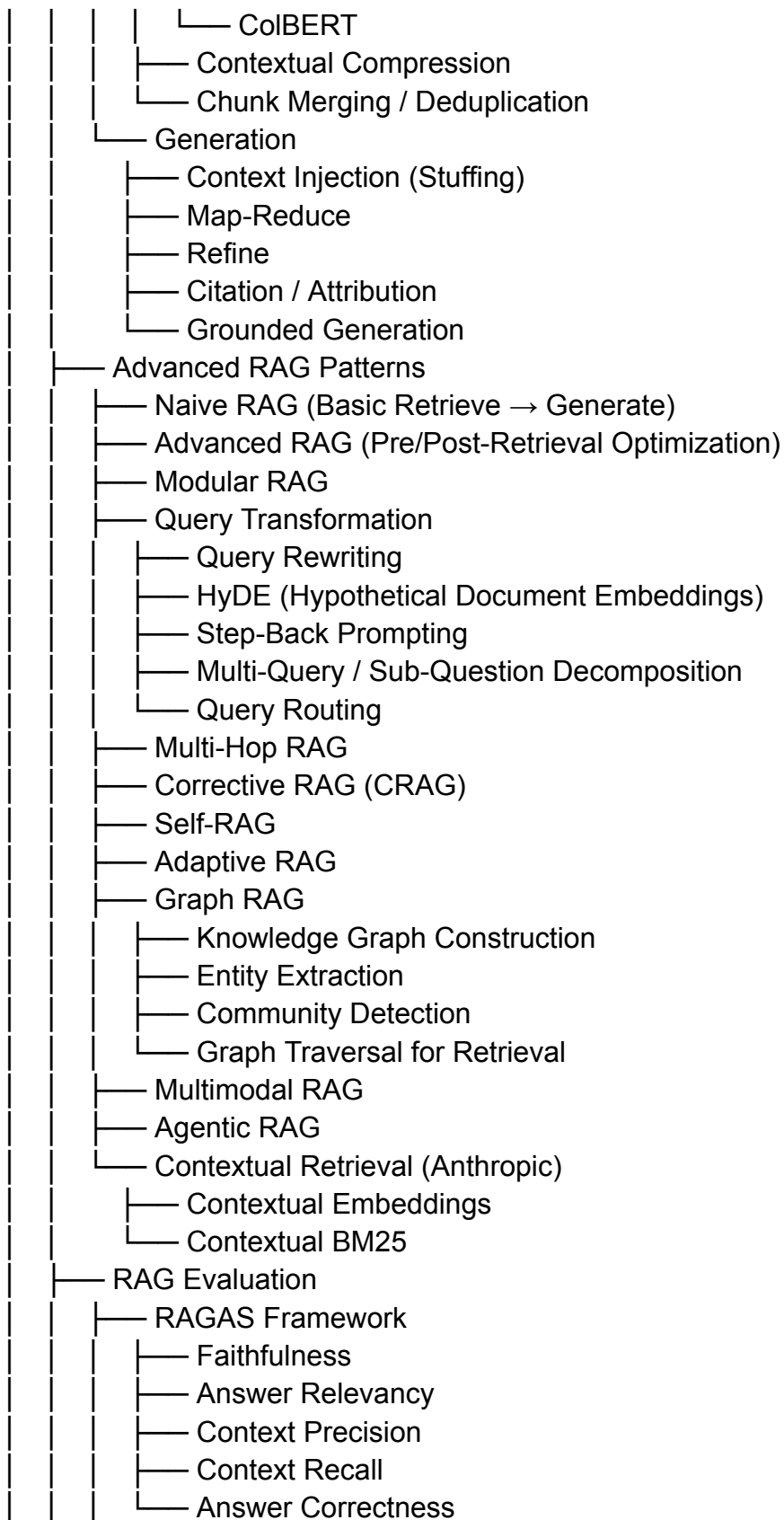
Memory Systems

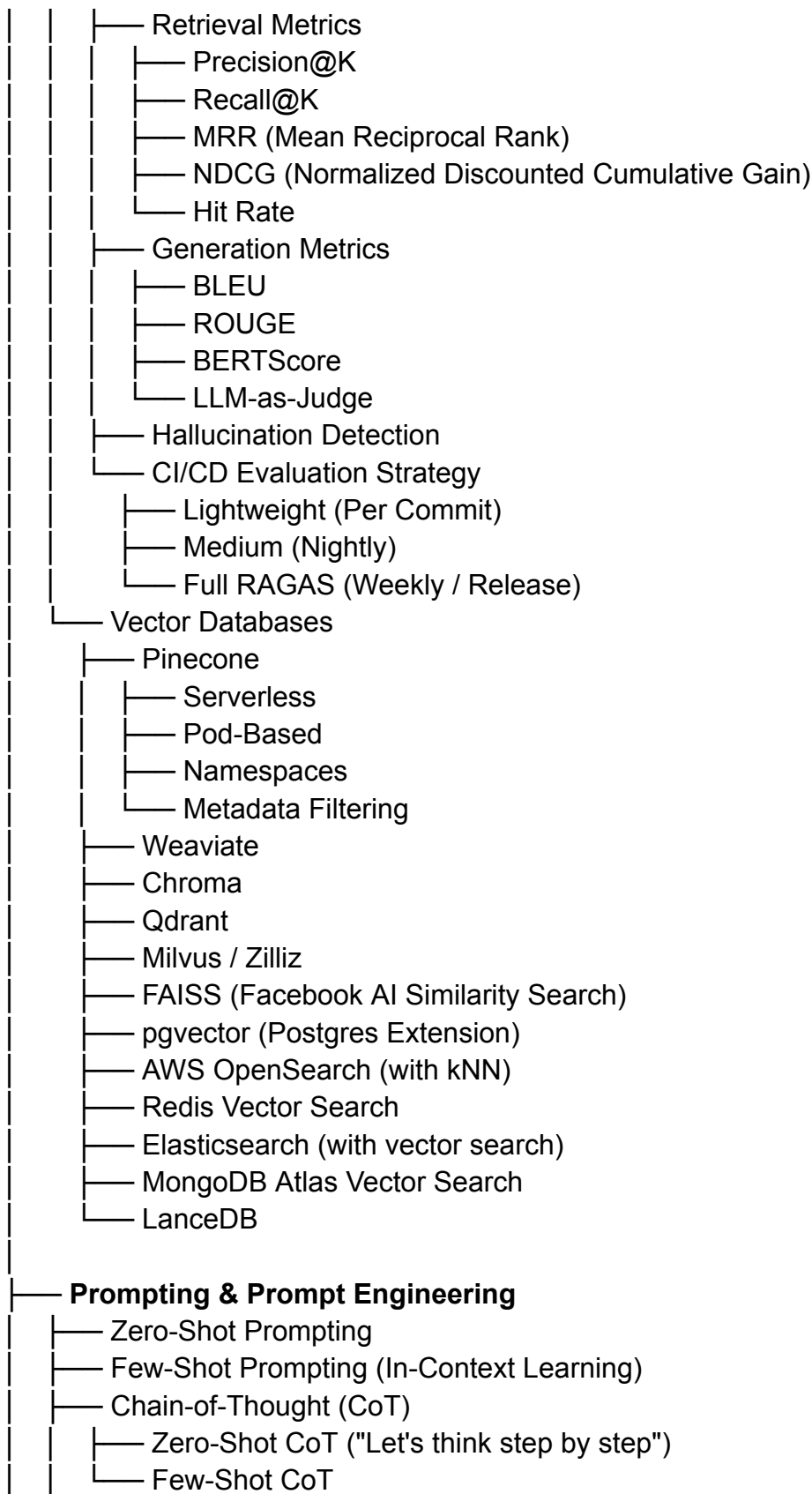












- Tree-of-Thought (ToT)
- Graph-of-Thought
- Self-Consistency
- ReAct Prompting
- System Prompt / System Message
- User Prompt
- Assistant Prompt (Pre-fill)
- Prompt Template
- Prompt Chaining
- Structured Output Prompting (JSON Mode)
- Role Prompting / Persona
- Instruction Tuning Alignment
- Meta-Prompting
- Prompt Injection (Attack)
- Prompt Leaking (Attack)
- Jailbreaking (Attack)

— **Fine-Tuning & Adaptation**

- Full Fine-Tuning
- Parameter-Efficient Fine-Tuning (PEFT)
 - LoRA (Low-Rank Adaptation)
 - QLoRA (Quantized LoRA)
 - Adapter Layers
 - Prefix Tuning
 - Prompt Tuning (Soft Prompts)
- RLHF (Reinforcement Learning from Human Feedback)
 - Reward Model
 - PPO (Proximal Policy Optimization)
 - Human Preference Data
- DPO (Direct Preference Optimization)
- RLAI (RL from AI Feedback)
- Constitutional AI (CAI)
- Instruction Tuning / SFT (Supervised Fine-Tuning)
- Continued Pre-Training / Domain-Adaptive Pre-Training
- Distillation
 - Knowledge Distillation
 - Teacher-Student Framework
 - Synthetic Data Generation for Distillation
- Merging
 - Model Merging (TIES, DARE, SLERP)

- └─ Frankenmerging
- └─ Training Data
 - └─ Data Curation
 - └─ Data Decontamination
 - └─ Synthetic Data Generation
 - └─ Data Augmentation

— **Safety & Guardrails**

- └─ Input Guardrails (Pre-processing)
 - └─ Prompt Injection Detection
 - └─ Topic Filtering
 - └─ PII Detection / Redaction
 - └─ Content Moderation
- └─ Output Guardrails (Post-processing)
 - └─ Hallucination Detection
 - └─ Toxicity Filtering
 - └─ Schema Validation
 - └─ Factual Grounding Check
 - └─ Sensitive Information Filtering
- └─ Guardrails Frameworks
 - └─ NeMo Guardrails (NVIDIA)
 - └─ Guardrails AI
 - └─ LLM Guard
 - └─ OpenAI Moderation API
- └─ Alignment
 - └─ RLHF
 - └─ Constitutional AI
 - └─ DPO
 - └─ Value Alignment
 - └─ Red Teaming
- └─ Responsible AI
 - └─ Bias Detection & Mitigation
 - └─ Fairness
 - └─ Transparency / Explainability
 - └─ AI Ethics
 - └─ Model Cards
- └─ Hallucination
 - └─ Intrinsic Hallucination (Contradicts source)
 - └─ Extrinsic Hallucination (Unverifiable claims)
 - └─ Mitigation Strategies (RAG, Grounding, Citations)

Security

Authentication & Authorization

- API Key Management
- OAuth 2.0 / OIDC
- JWT Tokens
- Role-Based Access Control (RBAC)
- Scoped Permissions (per tool / per agent)

Prompt Security

- Direct Prompt Injection
- Indirect Prompt Injection (via retrieved docs)
- Prompt Leaking
- Jailbreaking
- Adversarial Inputs

Data Security

- Data Encryption (at rest / in transit)
- PII Handling & Redaction
- Data Loss Prevention (DLP)
- Data Residency / Sovereignty

API Security

- Rate Limiting
- Input Validation & Sanitization
- Request Signing
- CORS / CSRF Protection

Secrets Management

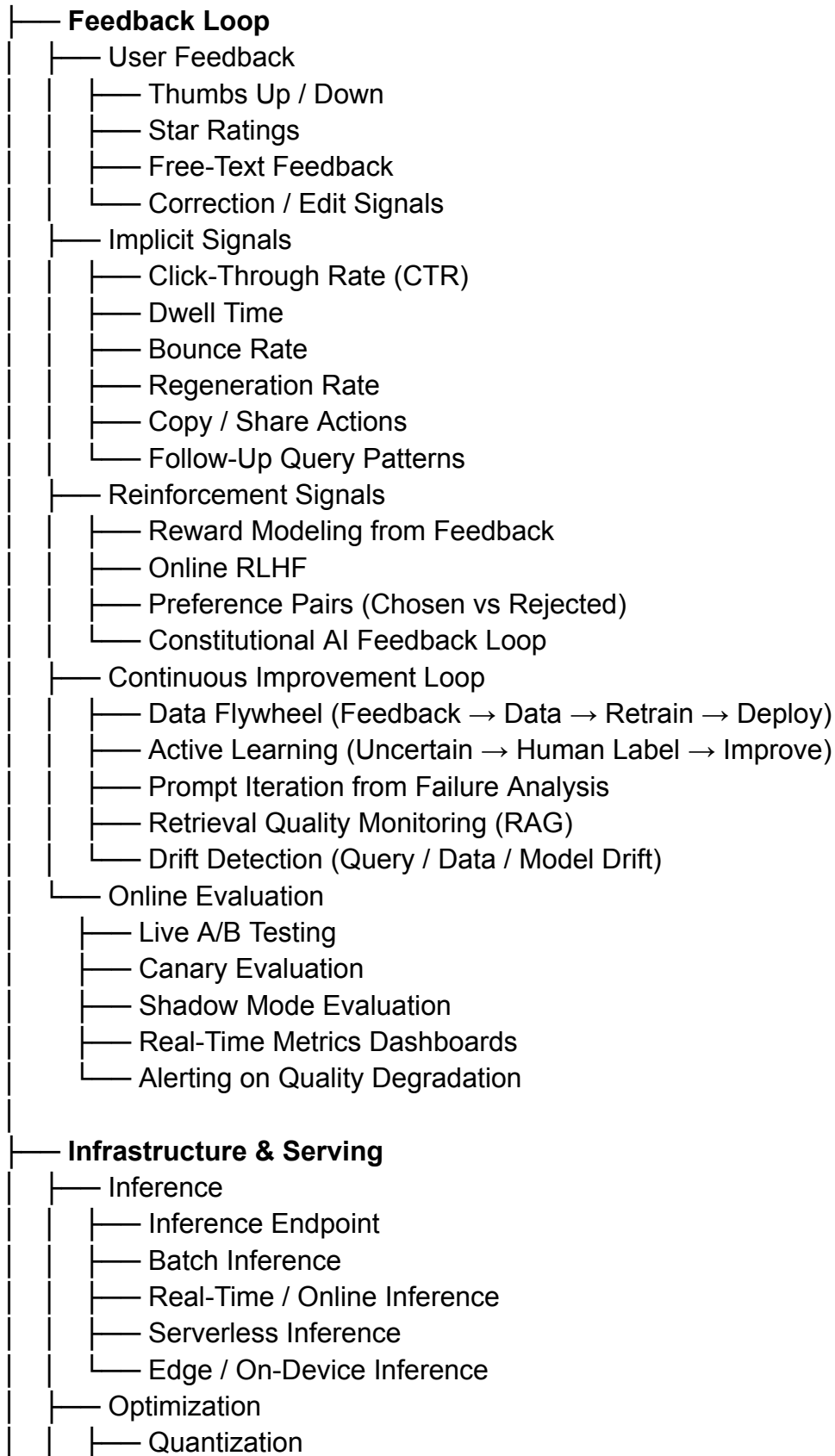
- Vault (HashiCorp)
- AWS Secrets Manager
- Environment Variables
- Key Rotation Policies

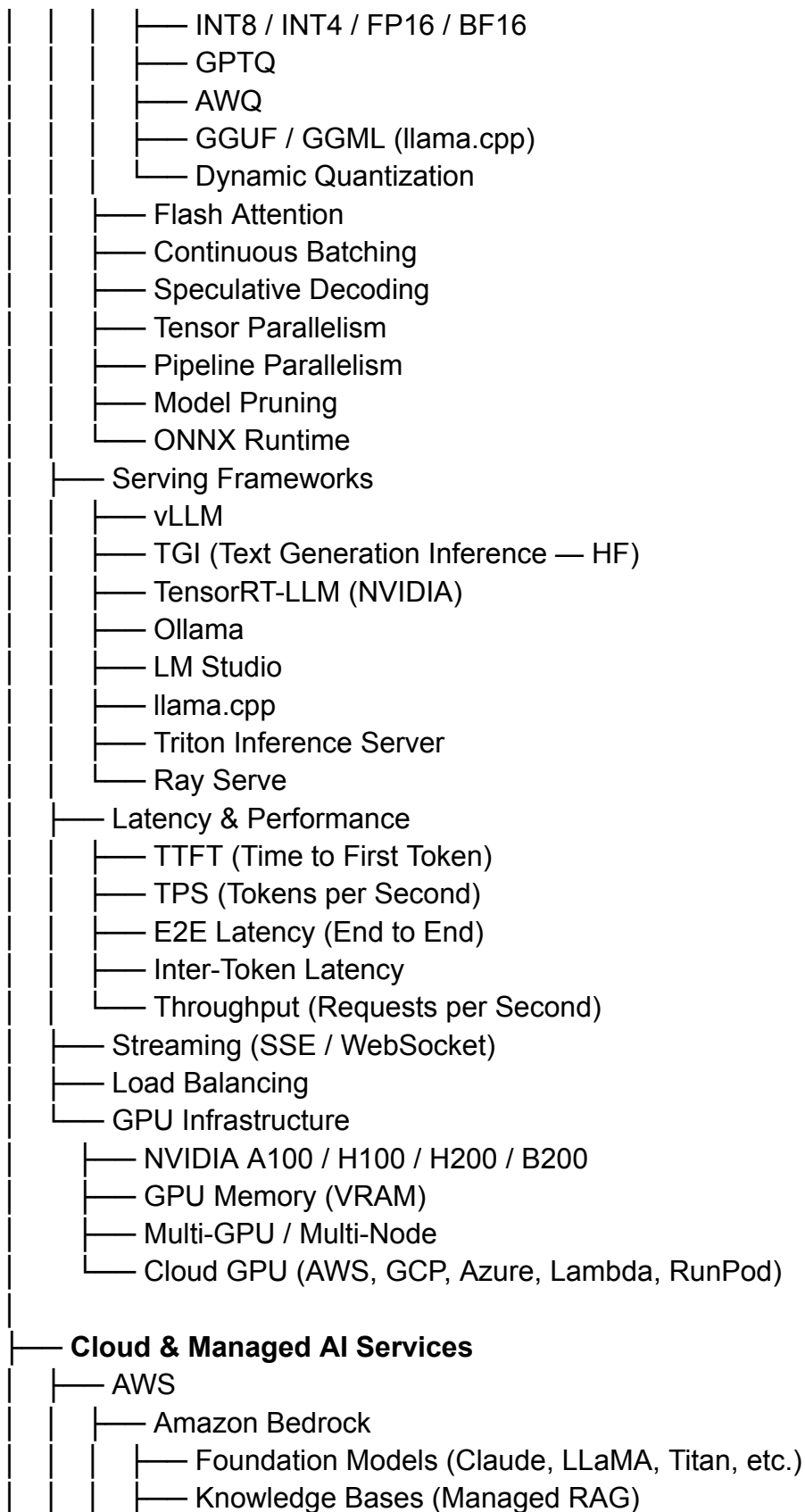
Supply Chain Security

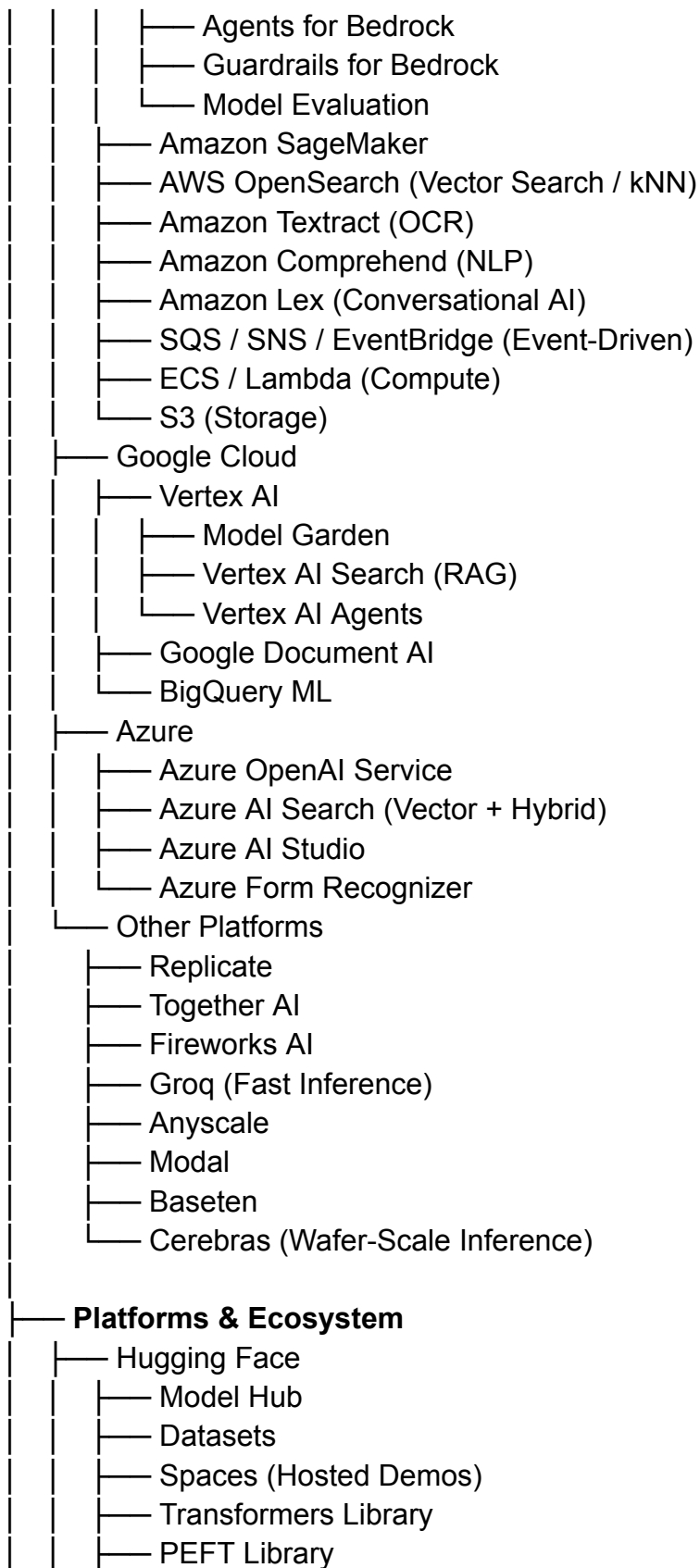
- Model Provenance
- Dependency Scanning
- Container Security

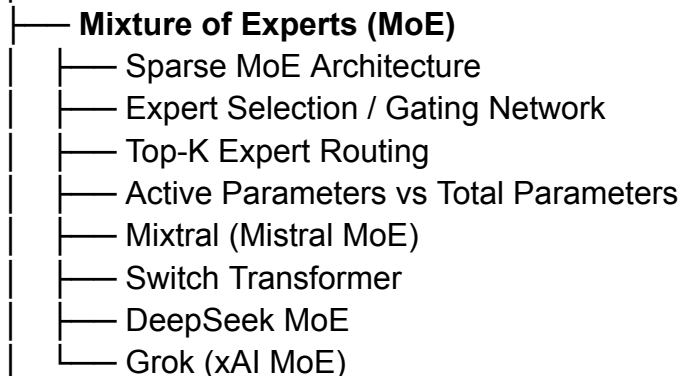
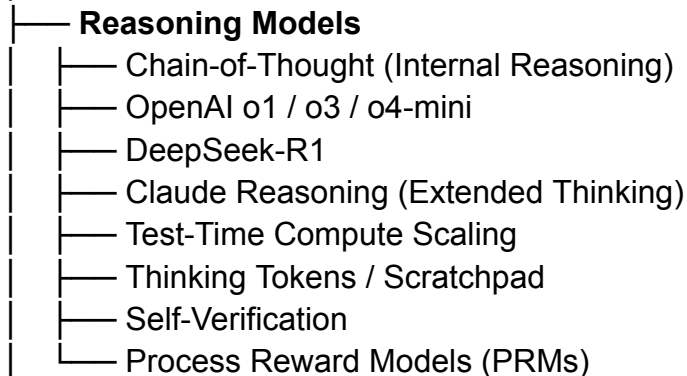
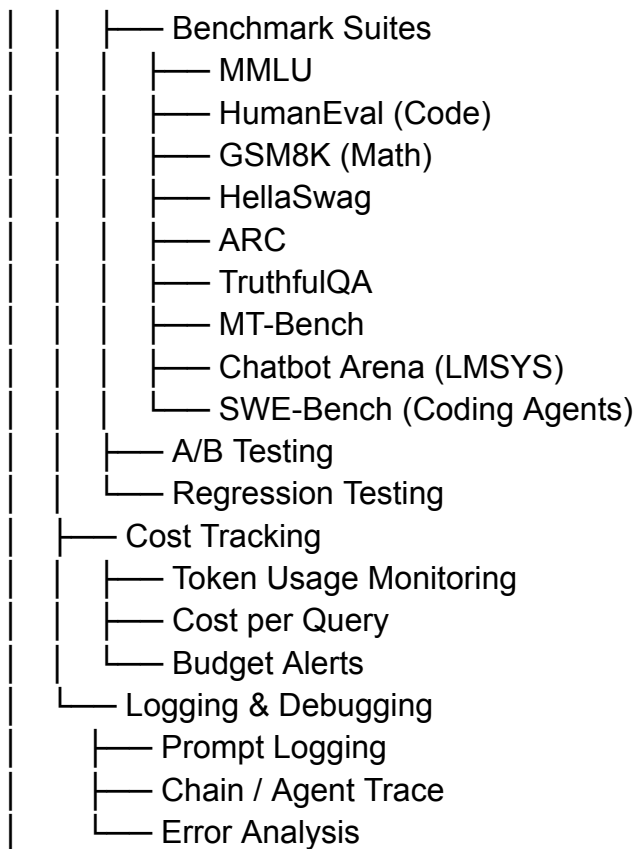
Audit & Compliance

- Audit Logging
- SOC 2 / ISO 27001
- GDPR / CCPA
- HIPAA









Structured Output

- JSON Mode
- JSON Schema Enforcement
- Tool Use / Function Calling (as structured output)
- Grammar-Constrained Decoding
- Pydantic (Python Validation)
- Instructor Library
- Outlines (Structured Generation)
- BAML

API Concepts

- Chat Completions API
- Messages API (Anthropic)
- Assistants API (OpenAI)
- System / User / Assistant Roles
- Streaming (SSE)
- Function Calling / Tool Use
- Vision API (Image Input)
- Batch API
- Embeddings API
- Moderation API
- API Key / Authentication
- Rate Limits (TPM, RPM)
- Pricing (Input/Output Tokens)
- SDK (Python, TypeScript, Java)
- Webhooks
- Idempotency

Image & Video Generation

- Diffusion Models
 - Stable Diffusion (Stability AI)
 - DALL·E (OpenAI)
 - Midjourney
 - Imagen (Google)
 - Flux
 - SDXL
- Key Concepts
 - Text-to-Image
 - Image-to-Image

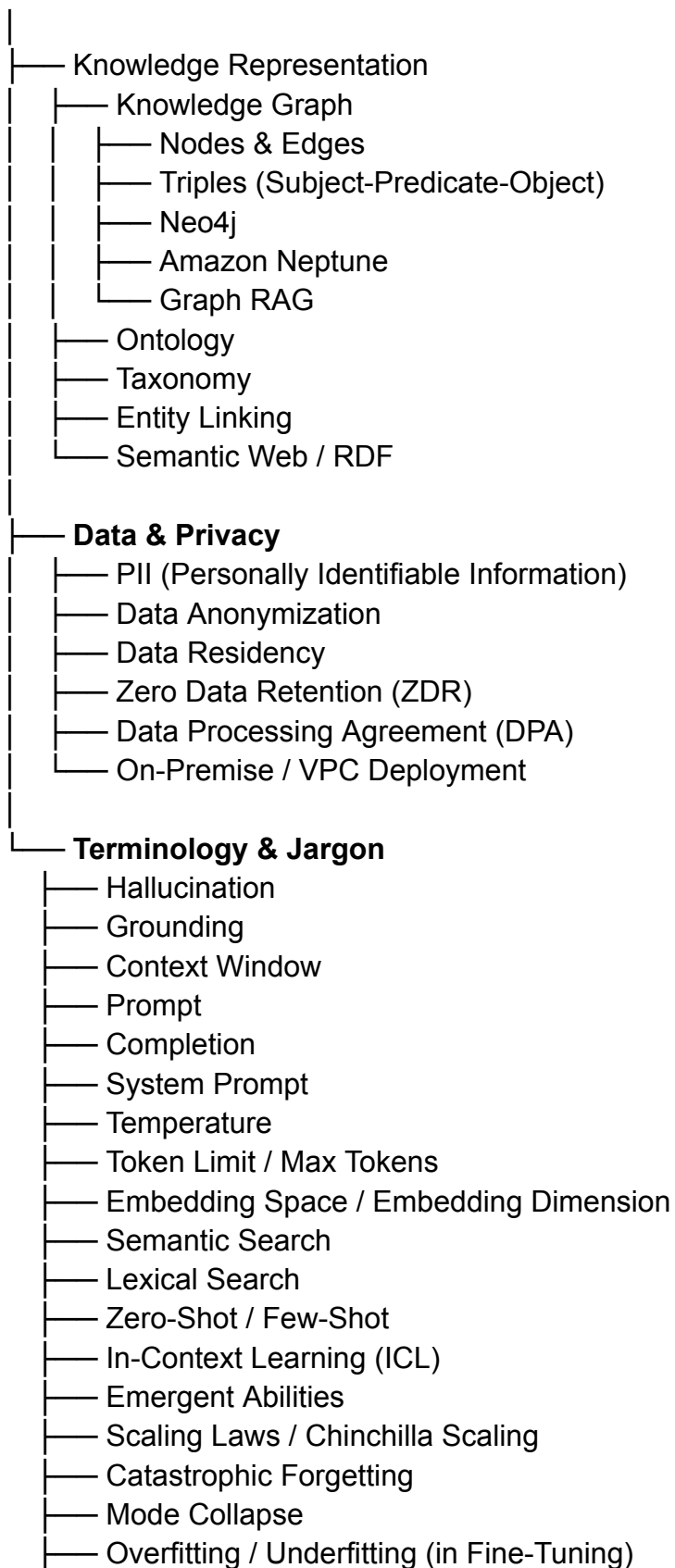
- Inpainting / Outpainting
- ControlNet
- Negative Prompt
- CFG Scale (Classifier-Free Guidance)
- Denoising Steps
- Latent Space
- Video Generation
 - Sora (OpenAI)
 - Runway Gen-3
 - Kling
 - Pika
 - Stable Video Diffusion
- Audio Generation
 - Text-to-Speech (TTS)
 - Voice Cloning (ElevenLabs)
 - Music Generation (Suno, Udio)
 - Sound Effects Generation

Coding Assistants & AI Dev Tools

- GitHub Copilot
- Claude Code (Anthropic)
- Cursor
- Windsurf (Codeium)
- Cody (Sourcegraph)
- Amazon CodeWhisperer / Q Developer
- Tabnine
- Aider
- Continue.dev
- Devin (Cognition)
- SWE-Agent

Conversational AI

- Chatbot
- Dialog Management
- Turn-Taking
- Context Carryover
- Slot Filling
- Intent Classification
- Multi-Turn Conversation
- Voice Assistants



- Epoch / Learning Rate / Batch Size
- Gradient Accumulation
- Loss Function / Warm-Up Steps
- Checkpoint / Model Registry / Model Versioning
- Canary / Shadow / Blue-Green Deployment
- Feature Flags (for LLM rollout)
- Evals / Vibes-Based Evaluation
- Ground Truth / Golden Dataset
- Annotation / Label / Active Learning
- Synthetic Data / Data Flywheel
- Compound AI System
- Foundation Model / Frontier Model
- SLM (Small Language Model)
- On-Device AI / Edge AI
- AI Gateway / LLM Proxy
- Latent Space / Representation Learning
- Transfer Learning / Domain Adaptation
- Cross-Lingual Transfer / Multilingual Models
- Code Generation / Code Completion
- Text-to-SQL / NL2Code
- Agentic Coding / Vibe Coding
- AI Pair Programming